

**VIDYAVARDHAKA COLLEGE OF ENGINEERING
MYSURU-570002**



**Department
Of
Computer Science & Engineering**

LABORATORY MANUAL

**“PYTHON PROGRAMMING LABORATORY”
(1BPLCSL207)**

VISION OF THE INSTITUTION

To be a globally renowned university imparting value-based education and promoting research to nurture responsible professionals for sustainable society.

MISSION OF THE INSTITUTION

1. To provide innovative teaching-learning through state-of-the-art infrastructure and competent staff.
2. To promote research and innovation in collaboration with industry and global institutions.
3. To inculcate professional ethics, leadership qualities, communication, and entrepreneurial skills for holistic development.

VISION OF THE DEPARTMENT

To establish a globally competent ecosystem in Computer Science and Engineering that nurtures future ready professionals through quality education and research for a sustainable society.

MISSION OF THE DEPARTMENT

1. To strengthen the technical expertise through industry-relevant curriculum, state-of-the-art infrastructure and dedicated staff.
2. To promote higher studies, research and innovation in collaboration with industry and academia.
3. To foster ethical values, leadership and entrepreneurial skills for holistic development.

PYTHON PROGRAMMING LABORATORY
(Effective from the academic year 2025 -2026)
SEMESTER – II

Course Name	: Python Programming Laboratory	Course Code : 1BPLCSL207
Number of Lecture Hours / Week	: 00	CIE Marks : 50
Number of Tutorial / Practical Hours / Week	: 02	SEE Marks : 50
Total Number of Lecture + Tutorial/Practical Hours	: 30	SEE Duration : 02 Hrs.
L:T:P	: 0:0:2	CREDITS : 01
COURSE PREREQUISITES: Basic knowledge of working with commands, logical thinking and problem solving skills are required to learn the course.		
COURSE OVERVIEW In this course the students will gain knowledge on the basic concepts of python syntax and semantics, core programming concepts like data structures, conditionals, loops, variables, writing custom functions, reading and writing to files. This course includes an overview of the various tools available for writing and running Python.		
COURSE LEARNING OBJECTIVES (CLO) of Python syntax and semantics to construct and utilize functions effectively for string processing and file organization ls to create and manipulate data structures such as lists and dictionaries in Python p solution for the given problem ncepts of Object-Oriented Programming as used in Python.		
PRACTICAL MODULE		TEACHING HOURS
<u>A- Demonstration</u> A1. Write a python program to convert temperature to and from Celsius to Fahrenheit. (Module- 1) A2. a) Develop a program to generate Fibonacci sequence of length (N). Read N from the console. (Module- 1) b) Write a function to calculate factorial of a number. (Module- 1) <u>B- Exercise</u> B1. Read a multi-digit number (as chars) from the console. Develop a program to print the frequency of each digit with suitable message. (Module- 2) B2. Read N numbers from the console and create a list. Develop a program to print mean, variance and standard deviation with suitable messages. (Module- 3) B3. Write a python program to sort dictionary elements based on key (Module- 3) B4. Write a python program to save only even numbers from a list to a file. (Module-3&4) B5. Write a python to create a list of words, group the words by their length and store it in a dictionary. (Module-3&4) B6. Write a function named DivExp which takes TWO parameters a, b and returns a value c (c=a/b). Write suitable assertion for a>0 in function DivExp and raise an exception for when b=0. Develop a suitable program which reads two values from the console and calls a function DivExp.(Module- 4) <u>C- Structured Inquiry</u> C1. Write a Python program to count frequency of characters in a given file. (Module- 4) C2. Write a program to read 3 subject marks and display pass or failed using class and object.(Module- 5) C3. Write a program to implement inheritance to inherit properties and functions of base class to derived class (Module- 5) <u>D- Open Ended Experiments</u> Design and develop solutions using Python Programming for problem statements present in hacker rank /codechef /leetcode		30

Textbooks	
1. Allen B. Downey, "Think Python: How to Think Like a Computer Scientist", 2nd Edition, Green Tea Press, 2015. (http://greenteapress.com/thinkpython2/thinkpython2.pdf) (Download pdf files from the above links)	
Reference Books	
1. Charles R. Severance, "Python for Everybody: Exploring Data Using Python 3", 1st Edition, CreateSpace Independent Publishing Platform, 2016. (http://do1.drchuck.com/pythonlearn/EN_us/pythonlearn.pdf)	
2. Gowrishankar S, Veena A, "Introduction to Python Programming", 1st Edition, CRC Press/Taylor & Francis, 2018. ISBN-13: 978-0815394372	
3. Reema Thareja, "Python Programming Using Problem Solving Approach", Oxford university press, 2017. ISBN-13: 978-0199480173	
Web links and Video Lectures (e-Resources)	
1. https://ocw.mit.edu/courses/6-0001-introduction-to-computer-science-and-programming-in-python-fall-2016/	
2. https://nptel.ac.in/courses/106106182	
COURSE OUTCOMES (COs)	
At the end of the course students will be able to	
CO1	Demonstrate problem solving skills using Python programming constructs
CO2	Analyze source code to identify and implement modifications that comply with specified requirements
CO3	Design and implement Python based solution for a given system requirement specification

CO – PO-PSO Matrix

CO	PO											PSO		
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3											2		
CO2		3										2		
CO3			2		2			2				2		
AVG.	3	3	2		2			2				2		

Conduct of CIE & SEE

1. All laboratory experiments are to be included for practical examination.
2. Students are allowed to pick one experiment from the lot and are given equal opportunity.
3. Change of experiment is allowed only once and marks allotted for writeup to be made zero.

Assessment Pattern:

CIE ASSESSMENT PATTERN

Components			Number of assessments	Marks Weightage	Max. Marks	CO weightage		
						CO1	CO2	CO3
(i)	CIE	Tests (A) [Individual]	2	40%	40	50%	50%	-
(ii)		OEE (B) [Group]	1	10%	10	-	-	100%
Total Marks					50			

Tests [20 marks]

Two tests will be conducted for 20 marks each. Total marks of 2 tests will be considered.

- Writeup – 4 marks
- Conduction – 12 marks
- Viva – 4 marks

SEE ASSESSMENT PATTERN

Semester End Examination (SEE) [50 marks]

- Writeup – 10 marks
- Conduction & results – 30 marks
- Viva – 10 marks

Components		Number of assessments	Marks Weightage	Max. Marks	CO weightage		
					CO1	CO2	CO3
(iii)	SEE - Semester end examination [<i>Individual</i>]	1	50%	50	50%	50%	-
Total Marks				50			

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EXPERIMENTS BASED ON SYLLABUS

Experiment A1

Write a python program to convert temperature to and from Celsius to Fahrenheit.

PROGRAM

```
c = float(input("Enter temperature in Celsius"))
# calculate fahrenheit
fahrenheit = (c* 1.8) + 32
print('%0.1f degree Celsius is equal to %0.1f degree Fahrenheit' %(c,fahrenheit))

f = float(input("Enter temperature in Fahrenheit"))
# calculate Celsius
celsius = (f - 32) / 1.8
print('%0.1f degree Fahrenheit is equal to %0.1f degree Celsius' %(f,celsius))
```

OUTPUT

```
Enter temperature in Celsius : 88
88.0 degree Celsius is equal to 190.4 degree Fahrenheit
Enter temperature in Fahrenheit : 70
70.0 degree Fahrenheit is equal to 21.1 degree Celsius
```


Experiment A2

- a) **Develop a program to generate Fibonacci sequence of length (N). Read N from the console.**

PROGRAM

```
N = int(input("Enter number of terms? "))

def recur_fibo(n):
    if n <= 1:
        return n
    else:
        return(recur_fibo(n-1) + recur_fibo(n-2))

# check if the number of terms is valid
if N <= 0:
    print("Plese enter a positive integer")
else:
    print("Fibonacci sequence:")
    for i in range(N):
        print(recur_fibo(i))
```

OUTPUT

```
Enter number of terms? 7
Fibonacci sequence:
0
1
1
2
3
5
8
```

b) Write a function to calculate factorial of a number.

PROGRAM

```
def factorial(x):  
    if x == 0 or x == 1:  
        return 1  
    else:  
        # recursive call to the function  
        return x * factorial(x - 1)  
# take input from the user  
num = int(input("Enter a number: "))  
# call the factorial function  
result = factorial(num)  
print("The factorial of", num, "is", result)
```

OUTPUT

```
Enter a number: 4  
The factorial of 4 is 24
```

Experiment B1

Read a multi-digit number (as chars) from the console. Develop a program to print the frequency of each digit with suitable message.

PROGRAM

```
s=input("Enter a Number:")
d={}
print("Frequency of each digit:")
for ch in s:
    d[ch]=d.get(ch,0)+1
print(d)
```

OUTPUT

```
Enter a Number:1123456
Frequency of each digit:
{'1': 2, '2': 1, '3': 1, '4': 1, '5': 1, '6': 1}
```

Experiment B2

Read N numbers from the console and create a list. Develop a program to print mean, variance and standard deviation with suitable messages.

PROGRAM

```
import numpy as np
lst = []
# number of elements as input
n = int(input("Enter number of elements : "))
print("Enter the elements of the list:")
# iterating till the range
for i in range(0, n):
    ele = int(input())
    lst.append(ele) # adding the element
print(lst)
# Calculating average using average()
print("Mean=", np.average(lst))
# Calculating variance using var()
print("Variance=", np.var(lst))
# Calculating standard deviation
print("Standard Deviation=", np.std(lst))
```

OUTPUT

```
Enter number of elements : 5
Enter the elements of the list:
1
2
3
6
4
[1, 2, 3, 6, 4]
Mean= 3.2
Variance= 2.96
Standard Deviation= 1.7204650534085253
```

Experiment B3

Write a python program to sort dictionary elements based on key.

PROGRAM

```
# empty dictionary
my_dict = {}

# number of elements
n = int(input("Enter number of elements: "))

# taking input from user
for i in range(n):
    key = input("Enter key: ")
    value = input("Enter value: ")
    my_dict[key] = value

# sorting dictionary based on keys
sorted_dict = dict(sorted(my_dict.items()))

# displaying results
print("Original Dictionary:", my_dict)
print("Dictionary sorted by keys:", sorted_dict)
```

OUTPUT

```
Enter number of elements: 5
Enter key: 3
Enter value: Apple
Enter key: 6
Enter value: Mango
Enter key: 2
Enter value: Banana
Enter key: 1
Enter value: Grapes
Enter key: 9
Enter value: Orange
Original Dictionary: {'3': 'Apple', '6': 'Mango', '2': 'Banana', '1': 'Grapes', '9': 'Orange'}
Dictionary sorted by keys: {'1': 'Grapes', '2': 'Banana', '3': 'Apple', '6': 'Mango', '9': 'Orange'}
```

Experiment B4

Write a python program to save only even numbers from a list to a file.

```
numbers = []

n = int(input("Enter number of elements: "))

for i in range(n):
    num = int(input("Enter a number: "))
    numbers.append(num)

file = open("even_numbers.txt", "w")

for n in numbers:
    if n % 2 == 0:
        file.write(str(n) + "\n")

file.close()

print("Even numbers saved to file")
```

OUTPUT

```
Enter number of elements: 7
Enter a number: 1
Enter a number: 2
Enter a number: 3
Enter a number: 4
Enter a number: 5
Enter a number: 6
Enter a number: 7
Even numbers saved to file
```

File output (even_numbers.txt)

```
2
4
6
```

Experiment B5

Write a python to create a list of words, group the words by their length and store it in a dictionary.

PROGRAM

```
# create list of words
words = []
n = int(input("Enter number of words: "))

for i in range(n):
    word = input("Enter a word: ")
    words.append(word)
# dictionary to store grouped words
word_dict = {}
# group words by length
for w in words:
    length = len(w)
    if length in word_dict:
        word_dict[length].append(w)
    else:
        word_dict[length] = [w]
# display result
print("Words grouped by length:")
print(word_dict)
```

OUTPUT

```
Enter number of words: 4
Enter a word: CAR
Enter a word: bus
Enter a word: tempo
Enter a word: train
Words grouped by length:
{3: ['CAR', 'bus'], 5: ['tempo', 'train']}
```

Experiment B6

Write a function named DivExp which takes TWO parameters a, b and returns a value c ($c=a/b$). Write suitable assertion for $a>0$ in function DivExp and raise an exception for when $b=0$. Develop a suitable program which reads two values from the console and calls a function DivExp.

PROGRAM

```
def DivExp(a, b):
    try:
        c = a / b
        return c
    except ZeroDivisionError:
        print("Division by Zero!")

# input from user
n1 = float(input("Enter first number: "))
n2 = float(input("Enter second number: "))

# function call
result = DivExp(n1, n2)

print("Result =", result)
```

OUTPUT

Enter first number: 9	Enter first number: 9
Enter second number: 0	Enter second number: 5
Division by Zero!	Result = 1.8
Result = None	

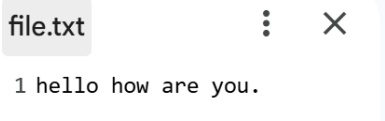
Experiment C1

Write a Python program to count frequency of characters in a given file.

PROGRAM

```
# file path
f = "/content/file.txt"
# open file in read mode
file = open(f, "r")
# list to store characters
a = []
# dictionary to store frequency
b = {}
# read file and add characters to list
for line in file:
    for char in line:
        a.append(char)
# count frequency of each character
for char in a:
    if char in b:
        b[char] += 1
    else:
        b[char] = 1
# close the file
file.close()
# print frequency dictionary
print(b)
```

OUTPUT

file.txt

1 hello how are you.

```
{'h': 2, 'e': 2, 'l': 2, 'o': 3, ' ': 4, 'w': 1, 'a': 1, 'r': 1, 'y': 1, 'u': 1, '.': 1}
```

Experiment C2

Write a program to read 3 subject marks and display pass or fail using class and object.

PROGRAM

```
class Stud:
    def display(self):
        print('Enter the Number of subjects:', end='')
        no_subj = int(input())

        for i in range(no_subj):
            subj = input('Enter the subject: ')
            marks = int(input('Enter the corresponding marks: '))
            if marks < 40:
                print(f'{subj}: Fail')
            else:
                print(f'{subj}: Pass')

        return self

# create object and call display
s = Stud()
s.display()
```

OUTPUT

```
Enter the Number of subjects:3
Enter the subject: Maths
Enter the corresponding marks: 70
Maths: Pass
Enter the subject: Physics
Enter the corresponding marks: 38
Physics : Fail
Enter the subject: Computer
Enter the corresponding marks: 90
Computer : Pass
```

Experiment C3

Write a program to implement inheritance to inherit properties and functions of base class to derived class.

PROGRAM

```
class Polygon:
    def findArea(self,base,height):
        return 0.5 *base *height

class Triangle(Polygon):
    pass

t = Triangle()

base = float(input("Enter the base of the triangle: "))
height = float(input("Enter the height of the triangle:"))
area = t.findArea(base,height)
print("The area of the triangle is:",area)
```

OUTPUT

```
Enter the base of the triangle: 7
Enter the height of the triangle:10
The area of the triangle is: 35.0
```

VIVA-VOCE QUESTIONS

1. What is a computer program?
2. What is Python?
3. How do you run a Python program?
4. What is the purpose of the print() function?
5. What are arithmetic operators in Python?
6. What is meant by values and types?
7. How do you check the type of a variable?
8. What is debugging?
9. What is a variable?
10. What rules should be followed while naming variables?
11. What is an assignment statement?
12. What is the difference between an expression and a statement?
13. What is script mode in Python?
14. What is operator precedence?
15. What are string operations?
16. What are comments and why are they used?
17. What is a function?
18. What is the difference between a fruitful and a void function?
19. What are parameters and arguments?
20. Why are functions used in programming?
21. What is floor division?
22. What is the modulus operator?
23. What is a Boolean expression?
24. What are logical operators?
25. What is conditional execution?
26. What is a chained conditional?
27. What is a nested conditional?
28. What is recursion?
29. What is a base case in recursion?
30. What is infinite recursion?
31. What is keyboard input in Python?
32. What is iteration?
33. What is variable reassignment?
34. What is a while loop?
35. What is the use of break?
36. What is a string?
37. How do you find the length of a string?
38. What does it mean that strings are immutable?
39. What is string slicing?
40. What is the use of string methods?
41. What is a list?
42. How are lists different from strings?
43. What does it mean that lists are mutable?
44. How do you traverse a list?
45. What are list operations?
46. What is list slicing?
47. Name some commonly used list methods.
48. What is the use of map()?
49. What is the use of filter()?
50. What is the use of reduce()?
51. How do you delete elements from a list?

52. What is aliasing?
53. How are lists passed as arguments to functions?
54. What is a dictionary?
55. How is a dictionary different from a list?
56. What is a key-value pair?
57. How do you loop through a dictionary?
58. What is a dictionary used as a counter?
59. What is reverse lookup?
60. What are memos in Python?
61. What is a tuple?
62. How are tuples different from lists?
63. What is tuple assignment?
64. Can a function return multiple values?
65. What are variable-length argument tuples?
66. How are lists and tuples related?
67. How are dictionaries and tuples used together?
68. Why are tuples used instead of lists?
69. What are random numbers in Python?
70. What is a word histogram?
71. What is meant by most common words?
72. What are optional parameters?
73. What is dictionary subtraction?
74. What is Markov analysis?
75. What is file handling?
76. How do you read a file in Python?
77. How do you write to a file?
78. What is exception handling in files?
79. What is pickling?
80. What is the format operator?
81. What is a class?
82. What is an object?
83. What are attributes?
84. What is an instance?
85. Are objects mutable in Python?
86. What is copying an object?
87. What is a method?
88. What is the purpose of the `__init__()` method?
89. What is the `__str__()` method?
90. What is operator overloading?
91. What is polymorphism?
92. What is inheritance?
93. What is a base class?
94. What is a derived class?
95. What are class attributes?
96. What is data encapsulation?
97. What is a class diagram?
98. What is type-based dispatch?
99. How is inheritance useful in real applications?
100. Give a real-world example of inheritance.